STEM SMART Single Maths 18 - Graph Transformations and Circles >

Graph Transformations

A-level Maths Topic Summaries - Functions

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

Fill in the blanks to complete the summary notes on graph transformations.

Part A

Graph transformations

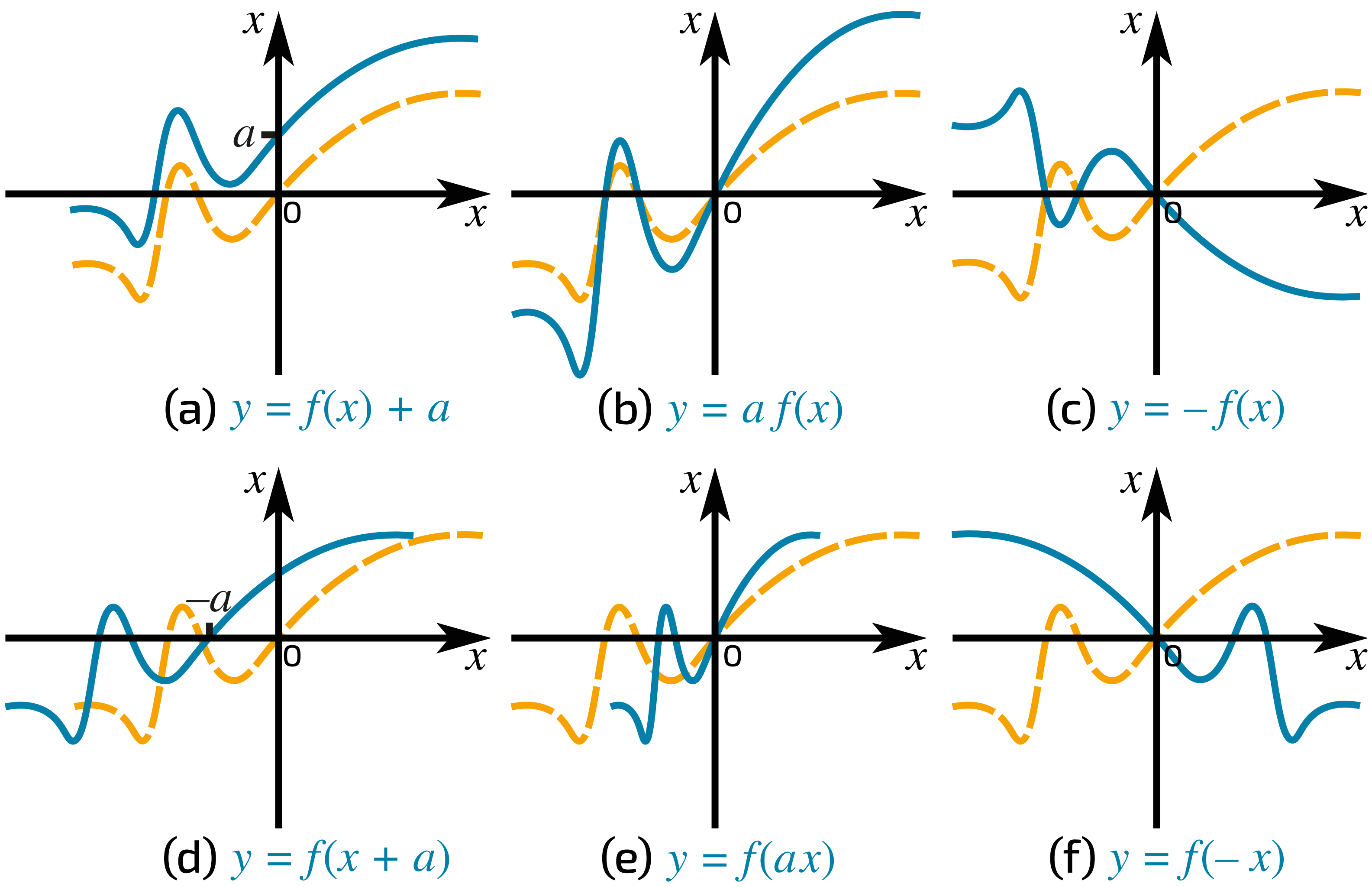

Figure 1: Transformations of the function $y = f(x)$. In each case the original function is the dashed yellow line, and the transformed function is the solid blue line.

Complete the table to describe the following transformations of the function $y = f(x)$.

Transformation	Description
$f(x) + a$	___ parallel to the ___ by ___
$af(x)$	___ parallel to the ___ by scale factor ___
$-f(x)$	___ in the ___
$f(x + a)$	___ parallel to the ___ by ___
$f(ax)$	___ parallel to the ___ by scale factor ___
$f(-x)$	___ in the ___

Items:

reflection | stretch | translation | x-axis | y-axis | $-a$ | $\frac{1}{a}$ | a

Part B

Order of transformations

If a function is translated and stretched parallel to the same axis, the order in which the transformations are carried out is important. This is illustrated in **Figure 2** and **Figure 3**.

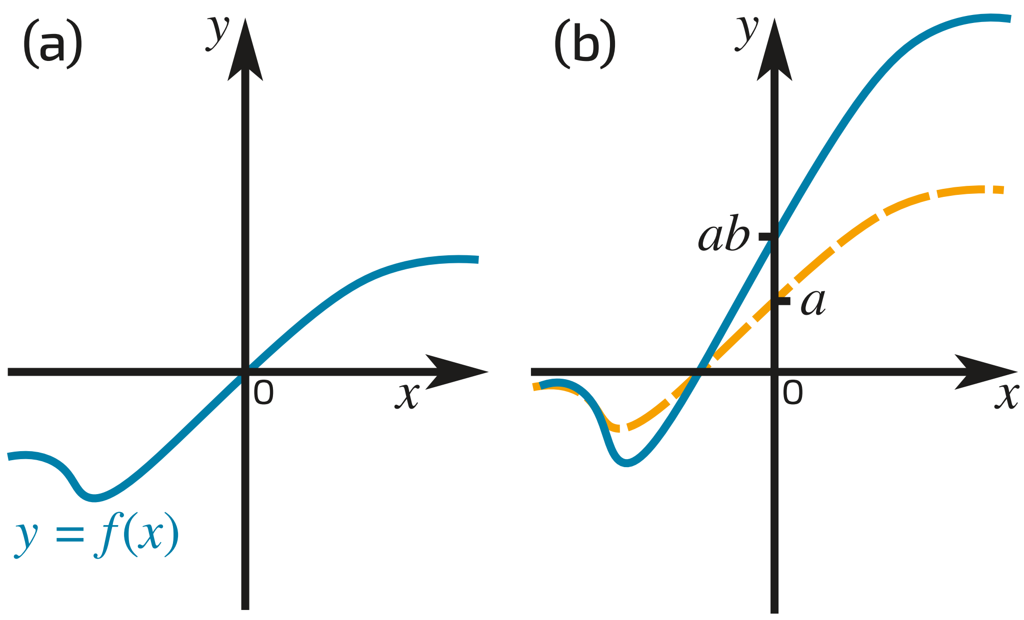

Figure 2: Transformations parallel to the y-axis. (a) Original function. (b) Translation by a (yellow dashed line) followed by stretch by b (blue line). (c) Stretch by b (yellow dashed line) followed by translation by a (blue line).

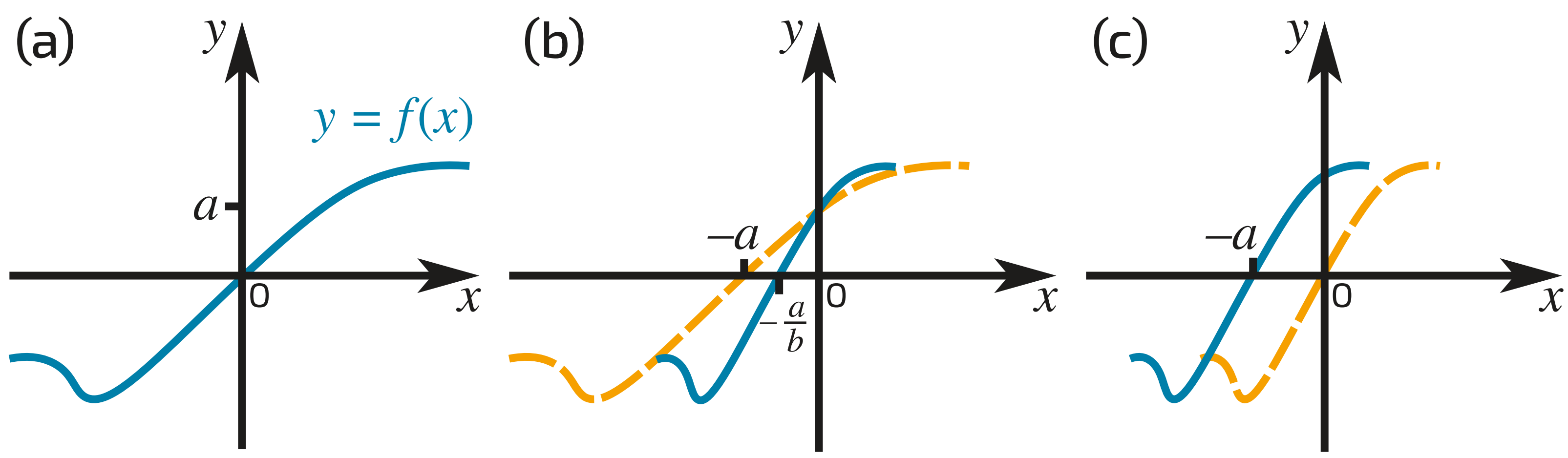

Figure 3: Transformations parallel to the x-axis. (a) Original function. (b) Translation by $-a$ (yellow dashed line) followed by stretch by $\frac{1}{b}$ (blue line). (c) Stretch by $\frac{1}{b}$ (yellow dashed line) followed by translation by $-a$ (blue line).

Use the diagrams above to help you complete these notes on transformation order.

Transformations parallel to the y-axis

Stretch by scale factor a followed by translation by b:

$$f(x) \to af(x) \to \boxed{}$$

Translation by b followed by stretch by scale factor a:

$$f(x) \to f(x) + b \to \boxed{}$$

Transformations parallel to the x-axis

Stretch by scale factor $\frac{1}{a}$ followed by translation by $-b$:

$$f(x) \to f(ax) \to \boxed{}$$

Translation by $-b$ followed by stretch by scale factor $\frac{1}{a}$:

$$f(x) \to f(x+b) \to \boxed{}$$

Items:

$f(ax+b)$ $f\ a(x+b)$ $af(x)+b$ $a\ f(x)+b$

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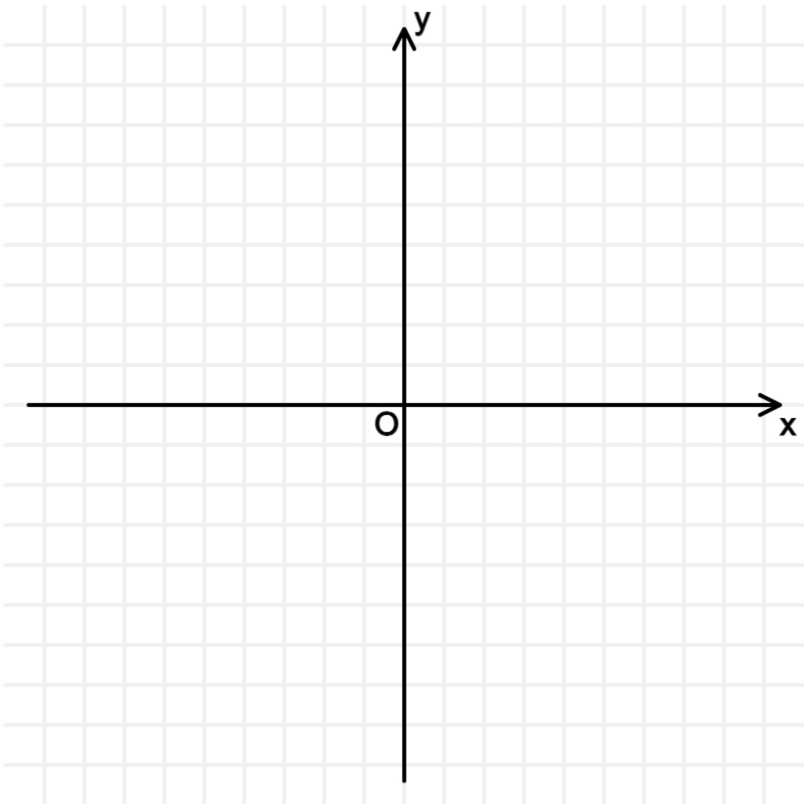
Transformations of Graphs 2ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

Part A

Sketch the curve $y = \frac{1}{x}$

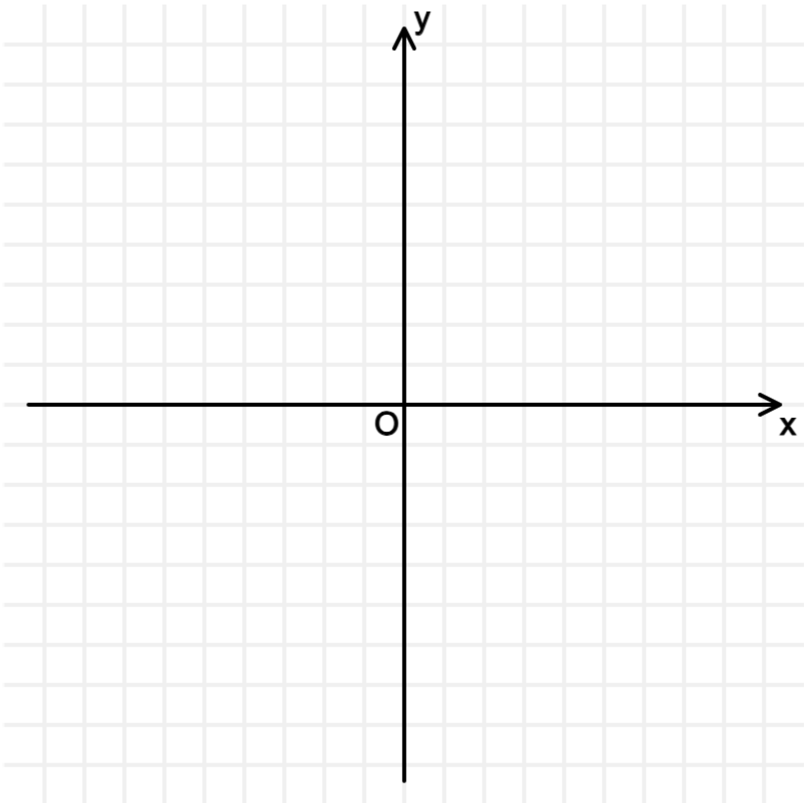
Sketch the curve $y = \frac{1}{x}$.



Part B

Sketch the curve $y = x^4$

Sketch the curve $y = x^4$.



Part C

Transformation from $y = x^3$ onto $y = 8x^3$

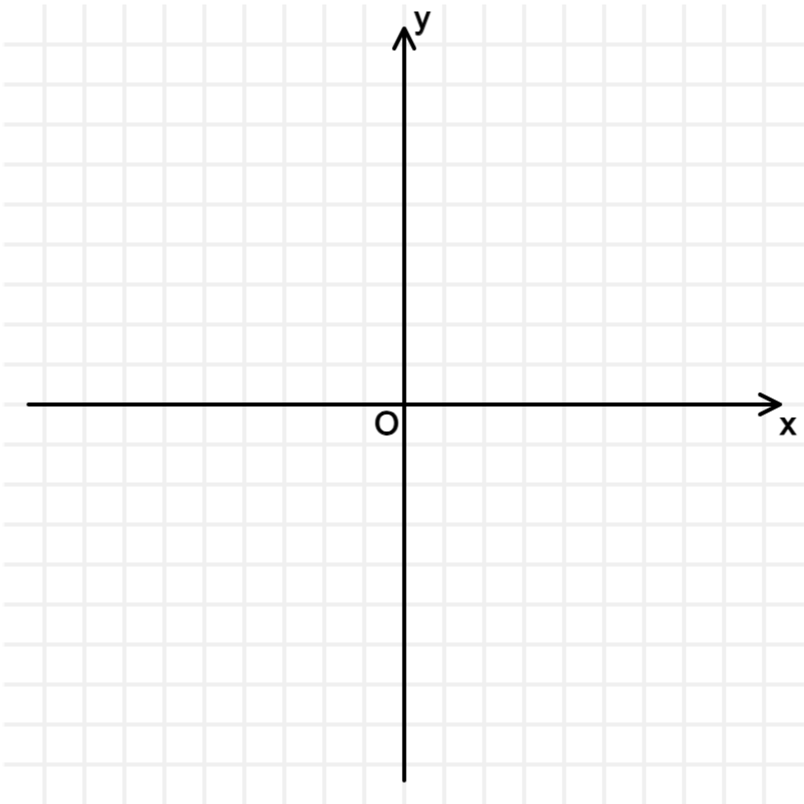
Which TWO of the following describe a single transformation that maps the curve $y = x^3$ onto the curve $y = 8x^3$?

- [ ] A stretch of scale factor 8 parallel to the y-axis
- [ ] A translation $+8$ units parallel to the y-axis.
- [ ] A stretch of scale factor $\frac{1}{8}$ parallel to the y-axis.
- [ ] A stretch of scale factor $\frac{1}{8}$ parallel to the x-axis.
- [ ] A stretch of scale factor $\frac{1}{2}$ parallel to the x-axis.
- [ ] A stretch of scale factor 8 parallel to the x-axis.

Part D

Sketch the curve $y = -\frac{1}{x}$

Sketch the curve $y = -\frac{1}{x}$.



Part E

State the equation

The curve $y = -\frac{1}{x}$ is translated by $+2$ units parallel to the x-axis in the positive direction. State the equation of the transformed curve.

The following symbols may be useful: `x`, `y`

Part F

Transformation from $y=-\frac{1}{x}$ onto $y=-\frac{1}{3x}$

Which TWO of the following describe a single transformation that maps the curve $y=-\frac{1}{x}$ onto the curve $y=-\frac{1}{3x}$?

- A stretch of scale factor $\frac{1}{3}$ parallel to the x-axis.
- A stretch of scale factor 3 parallel to the y-axis.
- A translation by $+3$ units parallel to the x-axis.
- A stretch of scale factor 3 parallel to the x-axis.
- A stretch of scale factor $\frac{1}{3}$ parallel to the y-axis.

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Transformations of Graphs 3ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

The graph of $y = f(x)$ for $-2 \leq x \leq 2$ is shown in **Figure 1**.

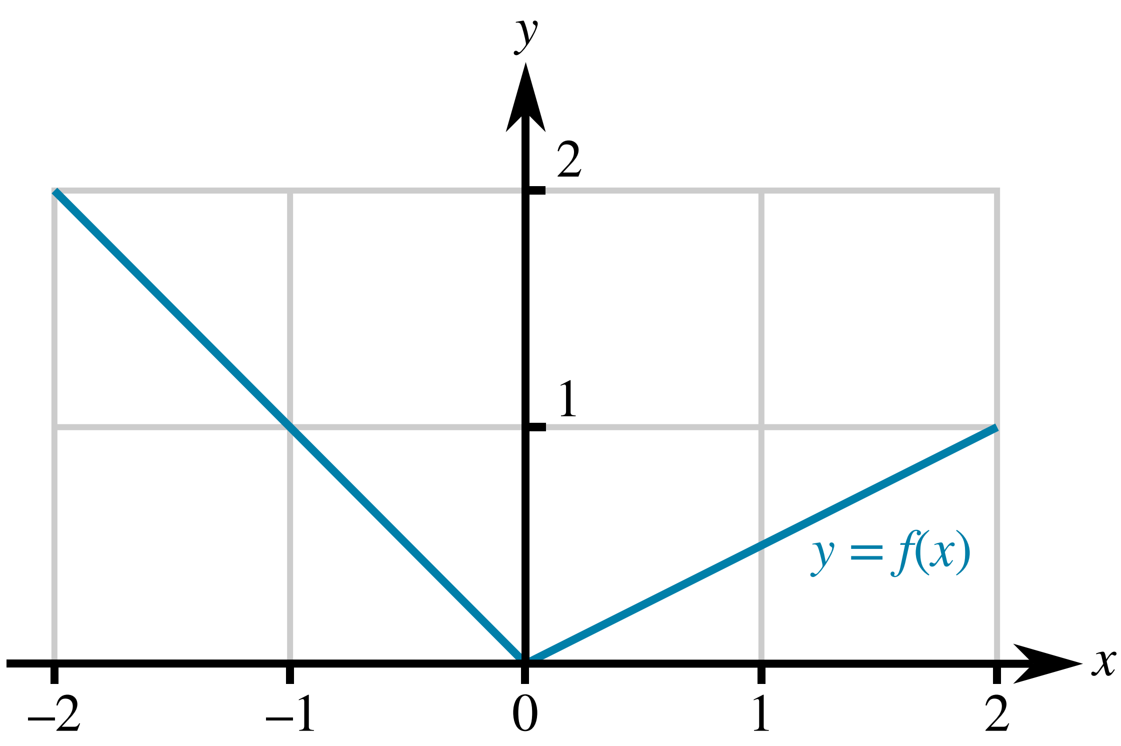


Figure 1: The graph of $y = f(x)$ for $-2 \leq x \leq 2$.

Part A

Sketch $y = f(-x)$

Sketch the curve $y = f(-x)$ for $-2 \leq x \leq 2$.

What is the y-value of the curve $y = f(-x)$ when $x = 1$?

The following symbols may be useful: y

Part B

Sketch $y = f(-x) + 2$

Sketch the curve $y = f(-x) + 2$ for $-2 \leq x \leq 2$.

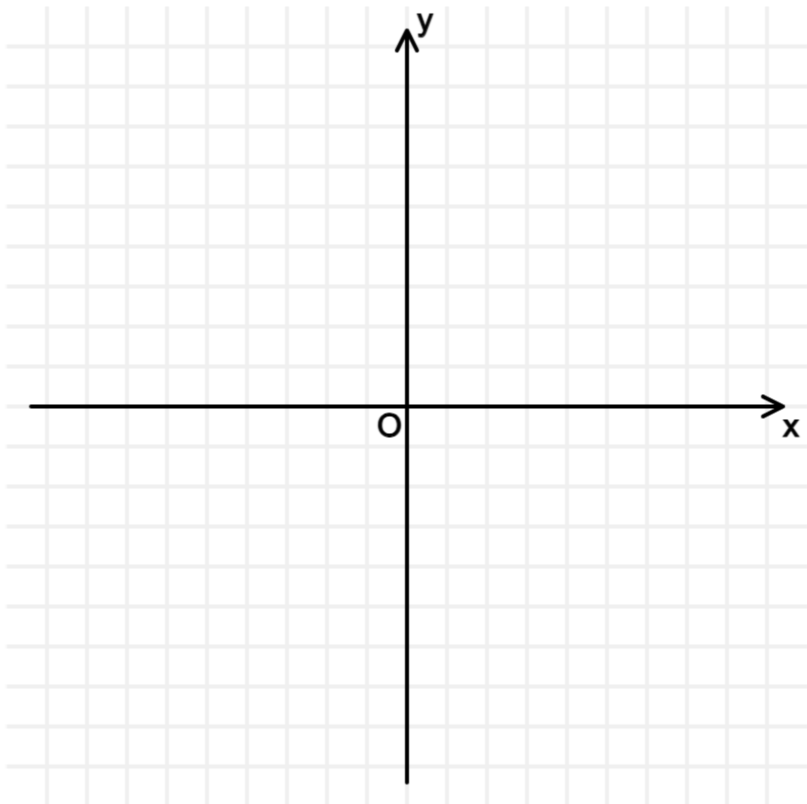
What is the y-value of the curve $y = f(-x) + 2$ when $x = -2$?

The following symbols may be useful: y

Part C

Sketch $y = -\frac{1}{x^2}$

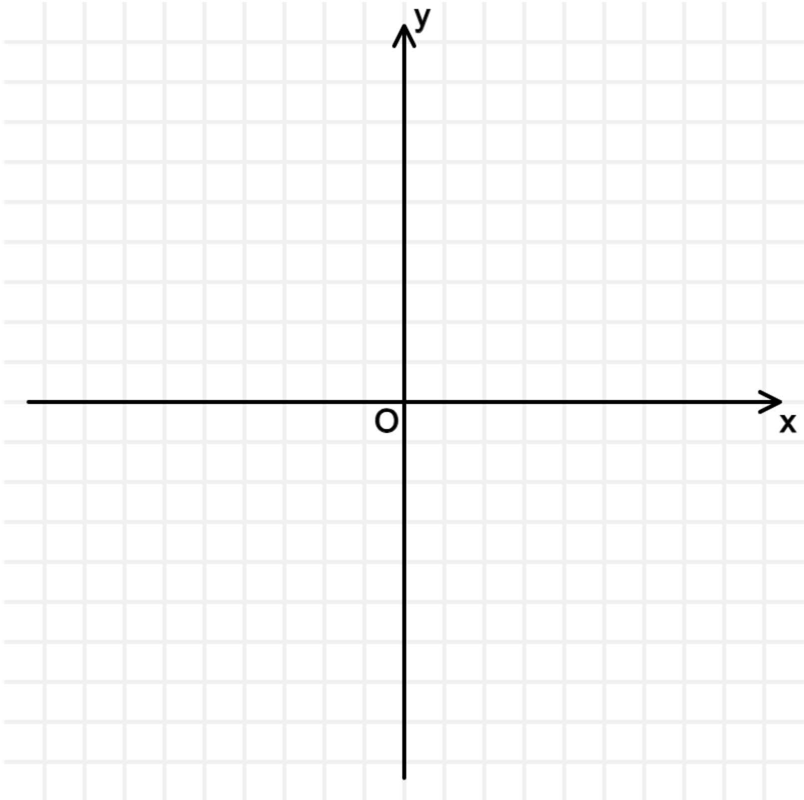
Sketch the curve $y = -\frac{1}{x^2}$.



Part D

Sketch $y = 3 - \frac{1}{x^2}$

Sketch the curve $y = 3 - \frac{1}{x^2}$.



Part E

State the equation

The curve $y = -\frac{1}{x^2}$ is stretched parallel to the y-axis by scale factor 2. State the equation of the transformed curve.

The following symbols may be useful: `x`, `y`

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Question

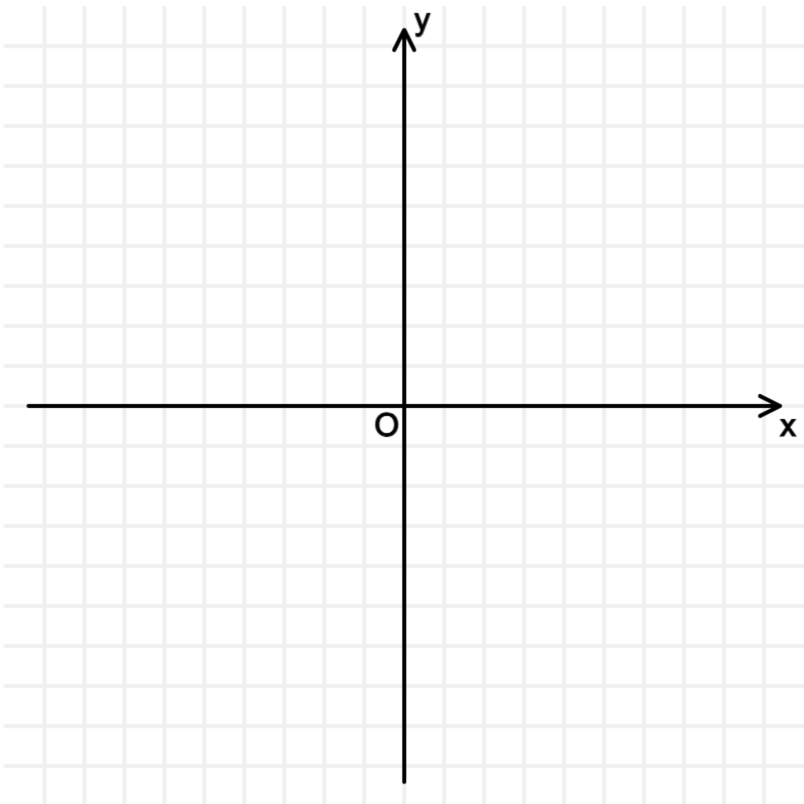
Transformations of Graphs 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

Part A

Sketch y

Find the roots of the curve $y = x^2(3 - x)$ and sketch it.



Part B

Translate y

The curve $y = x^2(3 - x)$ is translated by two units in the positive direction parallel to the x axis.

State the equation of the curve after this transformation.

The following symbols may be useful: `x`, `y`

Part C

Find transformation of y

Which of these describes the transformation of the curve $y = x^2(3 - x)$ to $y = \frac{1}{2}x^2(3 - x)$?

- A stretch of scale factor 2 parallel to the y-axis.
- A stretch of scale factor 2 parallel to the x-axis.
- A stretch of scale factor $\frac{1}{2}$ parallel to the y-axis.
- A stretch of scale factor $\frac{1}{2}$ parallel to the x-axis.

Part D

Vertical translation of $f(x)$

The curve $y = f(x)$ passes through the point P with coordinates $(2, 5)$.

State the coordinates of the point corresponding to P on the curve $y = f(x) + 2$.

(,)

Part E

Lateral stretching of $f(x)$

The curve $y = f(x)$ passes through the point P with coordinates $(2, 5)$.

State the coordinates of the point corresponding to P on the curve $y = f(2x)$.

(,)

Part F

Find transformation of $f(x)$

Which of the following describes the single transformation that maps the curve $y = f(x)$ onto $y = f(x + 4)$?

- A translation of -4 units parallel to the x-axis.
- A translation of -4 units parallel to the y-axis.
- A translation of 4 units parallel to the x-axis.
- A translation of 4 units parallel to the y-axis.

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Lateral and Vertical Translations

Pre-Uni Maths for Sciences E2.6

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

Investigate the transformations of the following functions.

Part A

Lateral translation

Consider the function $f(x) = x^2 + 2x + 1$. The function $g(x) = f(x - a)$, where a is a positive constant. If $g(1) = 9$, find the value of a.

If the value is not a whole number, enter the value as a decimal.

$a =$

Part B

Vertical translation

Consider the function $r(u) = \dfrac{2}{u-2}$. The function $s(u) = r(u) + b$, where b is a constant. If $s(0) = 1$, find the value of b.

If the value is not a whole number, enter the value as a decimal.

$b =$

Part C

Lateral and vertical translation

Consider the function $p(r) = \dfrac{1}{r}$. The function $q(r) = p(r - c) + d$, where c and d are constants. If $q(0) = 1$ and $q(2) = 3$, find the values of c and d.

If a value is not a whole number, enter the value as a decimal.

$c =$

$d =$

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isaac You work it out.

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Reflection and Symmetry

Pre-Uni Maths for Sciences E2.10

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level C2

The following questions ask you to deduce the symmetry properties of a number of functions. There are three choices:

even - a function for which $f(x) = f(-x)$ which is also described as being symmetric about the vertical axis,
odd - a function for which $f(x) = -f(-x)$ which is also described as being antisymmetric about the vertical axis (or symmetric about zero),
neither even nor odd.

Where relevant you may assume that a and b are non-zero constants.

Part A

Even functions

Decide which of the following functions are even.

- [ ] $a(x+b)^2$
- [ ] $(x-a)(x+a)$
- [ ] $a\cos x$
- [ ] $\frac{a}{x^2} + b$
- [ ] ax^2
- [ ] $ax^2 + bx^4$
- [ ] $ax^2 + b$
- [ ] $x^2(a+bx)$
- [ ] $a\sin x$
- [ ] $\frac{a}{x^2} + bx^2$
- [ ] $(x-a)(x+b)$ $(a \neq b)$

Part B

Odd functions

Decide which of the following functions are odd.

- [ ] $x(a+bx^2)$
- [ ] $x^{\frac{1}{3}}$
- [ ] $a\tan x$
- [ ] $\frac{a}{x}+bx^3$
- [ ] ax
- [ ] $x^2(a+bx)$
- [ ] $(x+a)^{\frac{1}{3}}$
- [ ] $\frac{a}{x}+b$
- [ ] $\frac{a}{x}$
- [ ] $\frac{a}{x}+\frac{b}{x^3}$
- [ ] $a\sin x$

Part C

Neither odd nor even functions

Decide which of the following functions are neither odd nor even.

- $(x-a)(x+a)^2$
- $(x-a)(x+a)$
- $ax-b$
- $ax^{\frac{1}{2}}$
- $x(ax^2+b)$
- $a\left(\frac{1}{x^2}-\frac{1}{b^2}\right)$
- $a(b-x)^{\frac{1}{2}}$
- $x^2(ax+b)$
- $\cos x+\sin x$
- $a\tan(x+45^\circ)$
- $\frac{a}{(x-b)^2}$

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isaac You work it out.

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Circles 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

The circle with equation $x^2 + y^2 - 6x - k = 0$ has radius 4.

The points A$(3, a)$ and B$(-1, 0)$ lie on the circumference of the circle, with $a > 0$.

Part A
Centre

By completing the square for x and y find the coordinates of the centre of the circle.

(,)

Part B
Value of k

Find the value of k.

The following symbols may be useful: k

Part C
Length AB

Calculate the length of AB, giving your answer in simplified surd form.

Part D

Equation

Find the equation of the line AB. Give your answer in the form $y = mx + c$.

The following symbols may be useful: `x`, `y`

Used with permission from UCLES, A level, June 2007, Paper 4721, Question 9.

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Circles 3ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

A circle has centre $(3, 1)$ and radius 5, and a line has equation $y = 2x$.

Part A

Circle equation

Write down the equation of the circle.

The following symbols may be useful: x, y

Part B

Intersection points

Find the coordinates of the points of intersection of the line and the circle.

(,)

Part C

Point on the line

Find the coordinates of the point on the line which is closest to the centre of the circle.

(,)

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Circles 2i

Subject & topics: Maths **Stage & difficulty:** A Level P1

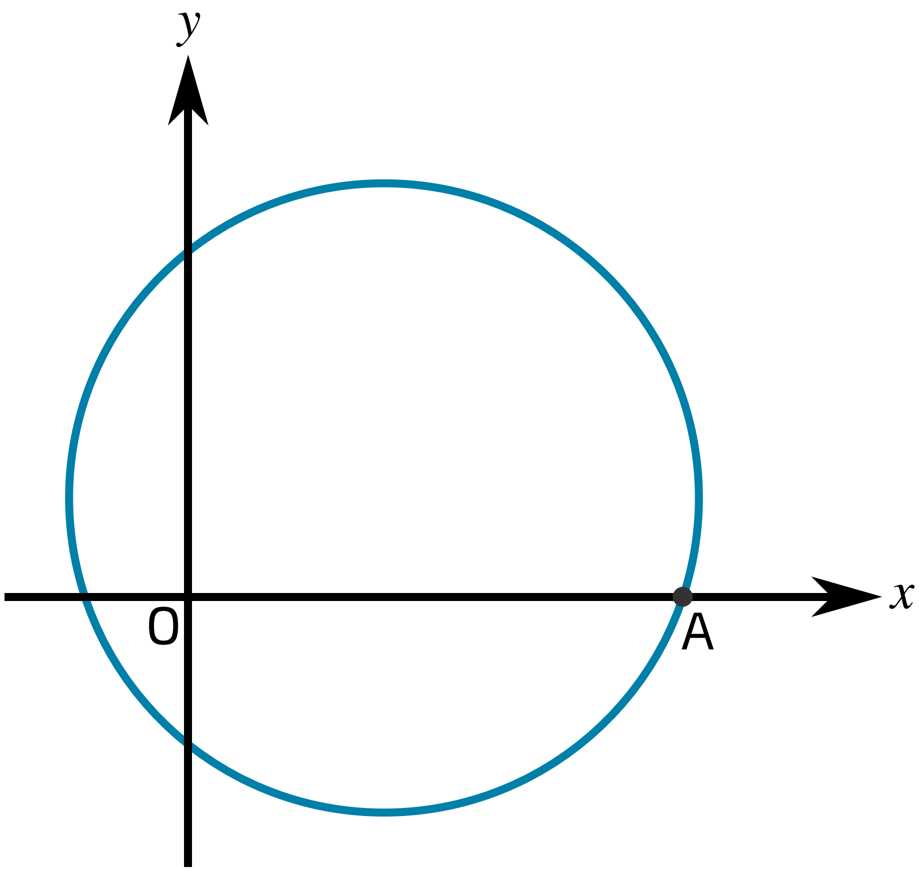

Figure 1: The circle with equation $x^2 + y^2 - 8x - 6y - 20 = 0$.

Figure 1 shows the circle with equation $x^2 + y^2 - 8x - 6y - 20 = 0$. The circle crosses the positive x axis at point A.

Part A

Find centre

By completing the square for x and y find the coordinates of the centre of the circle.

(,)

Part B

Find radius

Find the radius of the circle.

Part C

Tangent to the circle at A

Find the equation of the tangent to the circle at A. Give your answer in the form $y = mx + c$.

The following symbols may be useful: x, y

Part D

Another tangent to the circle

A second tangent to the circle is parallel to the tangent at A. Find the equation of this second tangent in the form $y = mx + c$.

The following symbols may be useful: x, y

Part E

Find a radius

Another circle has its centre at the origin O and radius r. This circle lies wholly inside the first circle and $r > 0$. Find the upper bound for r. Give your answer as an inequality.

The following symbols may be useful: `<`, `<=`, `>`, `>=`, `r`

Adapted with permission from UCLES, A level, June 2016, Paper 4721, Question 10.

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